14. AI Policy, Ethics, and Professional Practice

Rules, codes, and your responsibility as an engineer

Week 14 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on last week (Responsible AI)
2. Why AI needs rules — and the layers of rules
3. Regulation: the EU AI Act, the U.S. approach, and standards
4. Professional ethics: society codes, board rules, and the seal
5. IP, privacy, and integrity (summary; details are pre-reading)
6. Firm AI-use policies + debate: AI and shop-drawing review
7. Lab day: AI policy workshop · Lab 14 and HW5 due Sunday

Last week in 60 seconds

Week 13: what can go wrong, and how we manage it

1 Models break in new conditions

Distribution shift: a model trained on one climate, building type, or sensor can fail on another.

2 Bias hides in the data

Whose bridges, buildings, and neighborhoods are measured decides whom the model serves well.

3 Explain and quantify

Feature importance and calibrated uncertainty help, but explanations have limits too.

4 Manage risk on purpose

NIST AI RMF: Govern, Map, Measure, Manage. Document models with model cards.

Learning objectives

By the end of today, you can...

- ✓ Explain why AI needs rules, and name the layers: laws, board rules, standards, codes, policies
- ✓ Describe the EU AI Act's risk tiers and why critical-infrastructure AI is high-risk
- ✓ Contrast the EU and U.S. approaches, and name NIST AI RMF, ISO/IEC 42001, and 23894
- ✓ Tell society codes (NSPE, ASCE) from state board rules, and apply both to AI use and sealing
- ✓ Spot IP and data risks, and outline the nine parts of a firm's AI-use policy

01

Why policy?

Who answers when an AI system is wrong?

Why policy?

Who is responsible when the AI is wrong?

3 min

Think · Pair · Share

- Read the scenario. Alone, list **everyone** who might share responsibility
- With a partner: for each person or organization, what should they have done differently?
- Which rules — laws, contracts, licensing rules, codes of ethics — apply? Share one answer

Scenario: A city screens bridge decks with drone photos and an AI crack model. The model misses a growing crack. The consultant's report, signed by an engineer, rates the deck “fair.” Months later, concrete falls onto the road below. No one is hurt.

Why policy?

Six layers of rules shape how engineers use AI



Governments · mandatory: EU AI Act, state AI laws, privacy and copyright law

Your license · Arizona Board of Technical Registration (A.A.C. R4-30-301)

NIST, ISO · voluntary, unless a contract or law requires them

NSPE, ASCE · membership and guidance — not your license

Your firm, client, or university · conditions of your job or enrollment

You · when no rule covers the case yet

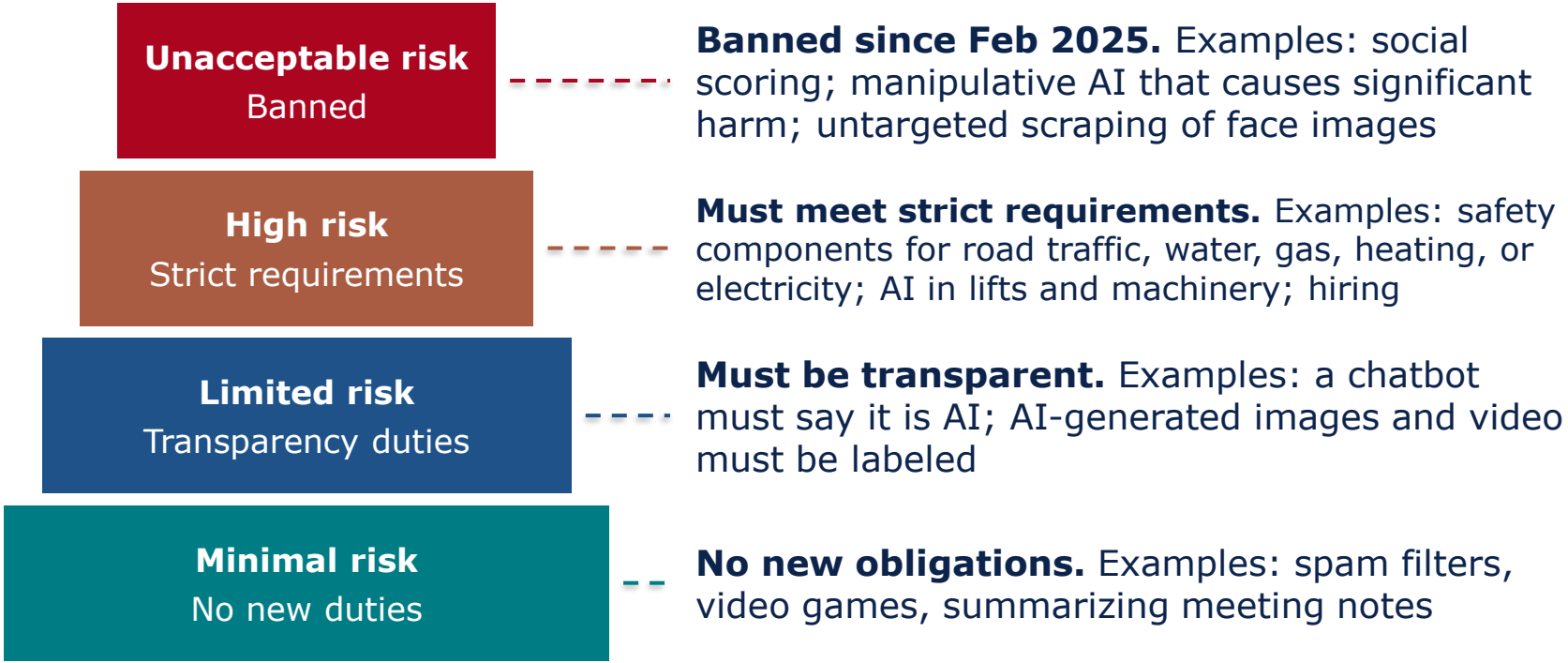
02

Regulation

The EU AI Act, the U.S. approach, and standards

Regulation

The EU AI Act regulates AI by risk: four tiers



Regulation

AI that runs critical infrastructure is high-risk

EU AI Act, Annex III, area 2 — Critical infrastructure

"AI systems intended to be used as safety components in the management and operation of critical digital infrastructure, road traffic, or in the supply of water, gas, heating or electricity."

Regulation (EU) 2024/1689, Annex III, point 2

- Could qualify: AI that operates traffic signals, chlorine dosing at a treatment plant, or valves and pumps in a water or district-heating network
- Key words: **safety component** and **intended use** — not the type of algorithm

Regulation

High-risk AI must meet six requirements before and during use

Articles 9–15 of the AI Act — then a conformity assessment before the system is sold

1 Risk management

Identify and reduce risks across the whole life cycle.

2 Data governance

Training data must be relevant, representative, and checked for errors and bias.

3 Documentation & logs

Technical documentation, plus automatic logs to trace problems later.

4 Information for users

Clear instructions: intended purpose, accuracy, and known limits.

5 Human oversight

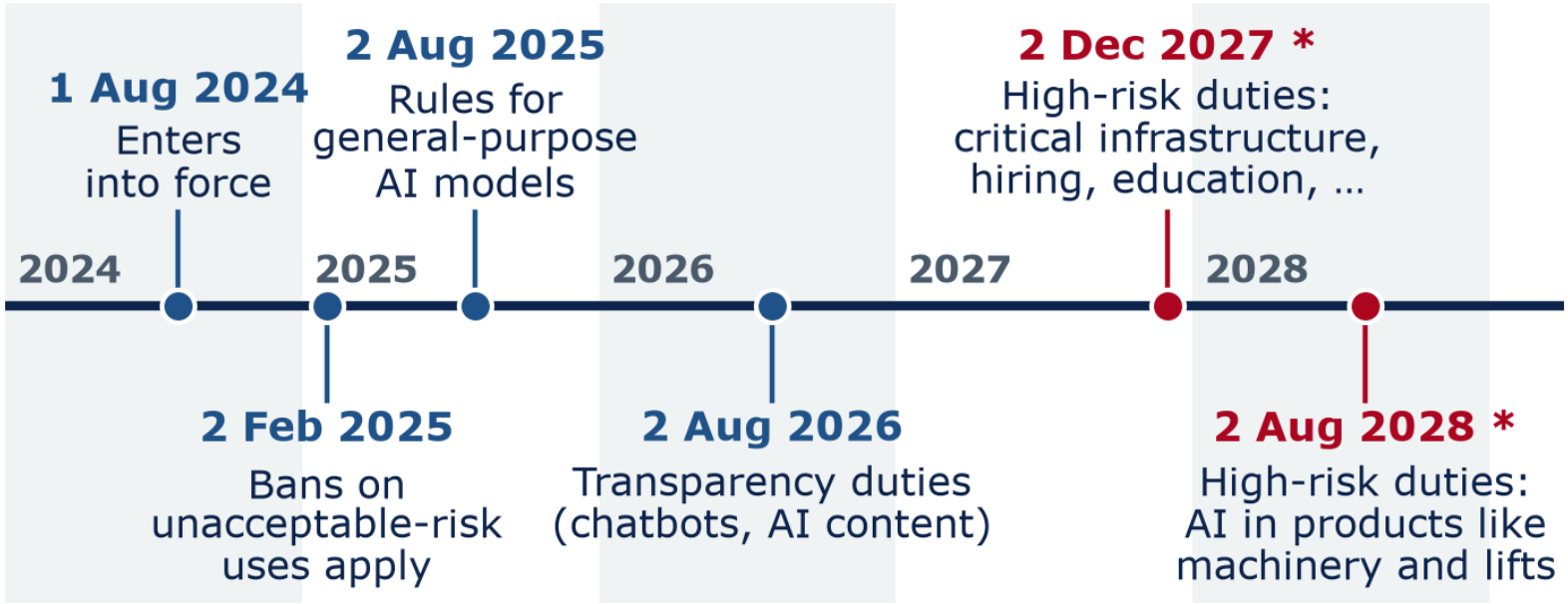
People can understand, monitor, override, or stop the system.

6 Accuracy & robustness

Tested performance; resilient to errors, faults, and cyberattacks.

Regulation

The AI Act's duties phase in over several years



* Moved later by a 2026 amendment, Regulation (EU) 2026/1744 (originally Aug 2026 and Aug 2027). The EU has changed these dates before — always check the current version.

Concept check

Spot the high-risk use

Under the EU AI Act, which use is most likely classified as high-risk?

A A spam filter on an engineering firm's email

B An AI system that is a safety component controlling chlorine dosing at a water-treatment plant

C A chatbot that answers permit questions on a city website

D A tool that suggests furniture layouts for an office floor plan

Concept check

Spot the high-risk use — answer

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Safety components in the supply of water are listed as high-risk (Annex III). The chatbot has transparency duties; A and D are minimal risk.

Regulation

The EU has one AI law; the U.S. has a patchwork

European Union

- One broad law (the AI Act, 2024) for all sectors
- Risk tiers set the duties; some uses banned
- Binding rules with large fines
- Reaches non-EU firms whose AI is used in the EU

vs

United States (as of 2026)

- No single federal AI law
- Sector regulators apply existing law (FTC, FDA, NHTSA…)
- A growing patchwork of state AI laws
- Federal AI policy shifted in 2025 — check current status

Either way, existing law still applies: negligence, contracts, privacy, and state licensing law

Voluntary standards turn good intentions into a management system

Framework	Type	What it gives you	Engineering analogy
NIST AI RMF 1.0 (2023)	U.S. government; voluntary; free	Four functions — Govern, Map, Measure, Manage — plus a generative-AI profile (2024)	A project risk plan
ISO/IEC 42001:2023	International standard; organizations can be certified	Requirements for an AI management system: policies, roles, controls, audits	ISO 9001 quality management — for AI
ISO/IEC 23894:2023	International guidance	How to apply risk management (ISO 31000) to AI systems	A risk register for AI

03

Professional ethics

Society codes, board rules, and the seal

Professional ethics

The NSPE canons already cover AI use

#	NSPE canon — engineers shall…	What it means when you use AI
1	Hold paramount the safety, health, and welfare of the public	Public safety outranks speed, cost, and convenience — including AI shortcuts
2	Perform services only in areas of their competence	AI does not extend your competence: you must be able to check its answer
3	Issue public statements only in an objective and truthful manner	Don't overstate what an AI-based analysis shows
4	Act for each employer or client as faithful agents or trustees	Protect client data: no confidential files in unapproved tools
5	Avoid deceptive acts	Don't misrepresent who — or what — did the work
6	Conduct themselves honorably, responsibly, ethically, and lawfully…	Follow laws, licenses, and tool terms; own your mistakes

Professional ethics

The ASCE Code says the same, in its own words

ASCE Code of Ethics (updated 2020), quoted from asce.org — engineers…

SOCIETY

Public safety

"first and foremost, protect the health, safety, and welfare of the public;"

CLIENTS AND EMPLOYERS

Competence

"perform services only in areas of their competence;"

CLIENTS AND EMPLOYERS

Sealing

"approve, sign, or seal only work products that have been prepared or reviewed by them or under their responsible charge."

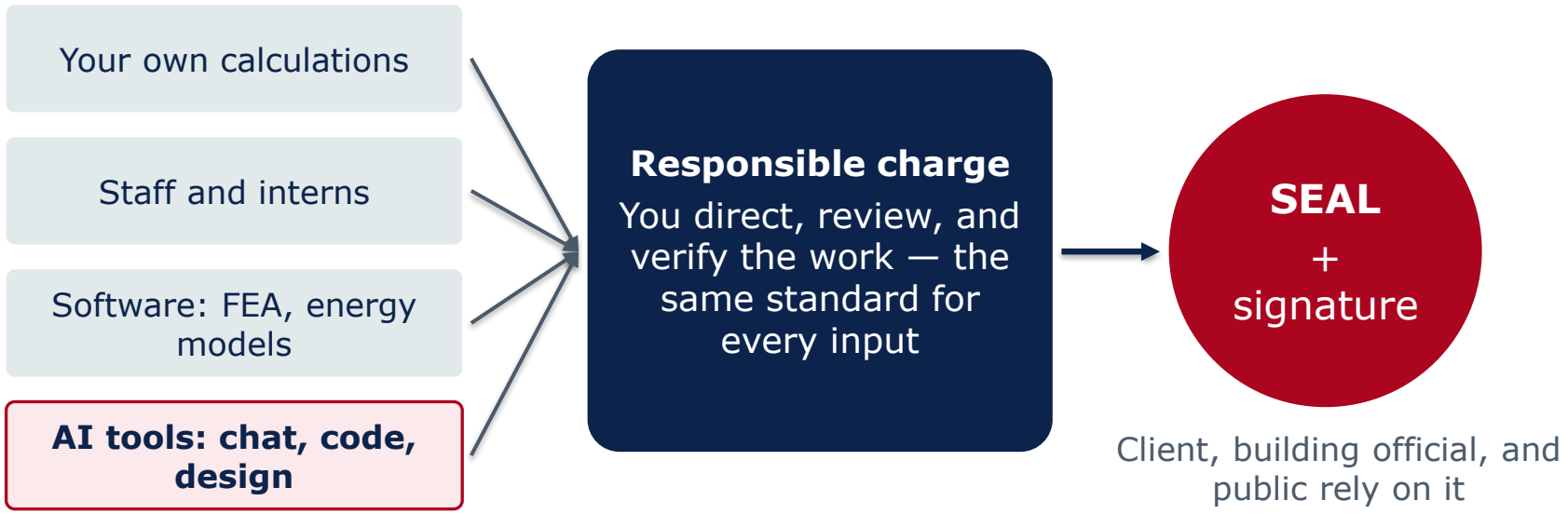
CLIENTS AND EMPLOYERS

Confidentiality

"keep clients' and employers' identified proprietary information confidential;"

Professional ethics

The seal means: “I am responsible for this work”



The tool is never the engineer of record. AI changes **how** work is made — not **who answers** for it.

Professional ethics

Engineering societies have published AI positions

ASCE Policy Statement 573 (2024)

- AI matters to civil engineering — and it is a tool
- The civil engineer stays responsible for planning, design, construction, operation, maintenance
- AI cannot replace a licensed PE's professional judgment
- Use AI under a licensed PE's direction; public safety comes first

NSPE Position Statement 03-1774 (adopted 2023, rev. Feb 2026)

- People who build or oversee AI that affects public safety should meet licensure-level standards
- Rigorous verification, validation, testing, and monitoring
- Transparency, accountability, and privacy protection
- Continuing education on AI for professionals

Concept check

Which data class — and what is the problem?

A project manager's AI note-taker records a site walk at a client's water plant. The transcript names plant staff and describes a weak point in the plant's control system. The vendor stores it. Best answer?

A Internal — fine, because the firm made the recording

B Restricted — personal and security details; it needs an approved tool and consent to record

C Public — the site walk happened outdoors

D Confidential (client) — no extra steps, since note-takers are common tools

Concept check

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Security details and names make it Restricted — the highest class in the file wins. It needs an approved tool, the client's agreement, and consent to record.

IP, privacy, and integrity

Ownership, data, and integrity: three rules to remember

Pre-reading: appendix slides A1–A3 (after Resources) give the details

© Copyright needs a human author

Material generated wholly by AI is not protected (U.S. Copyright Office, 2025). Contracts decide much of the rest.

! Classify data before you paste

Public · Internal · Confidential (client) · Restricted (personal and security-sensitive). Each class has its own tool rules.

✓ Our AI labels mirror practice

AI-Free ≈ licensure exams; AI-Assisted ≈ normal work; AI-Required ≈ testing a tool. UA's Code of Academic Integrity covers AI use.

04

Firm policy and debate

Writing the rules — and testing them

A firm AI-use policy has nine parts — Lab 14 uses the same list

§	Section	The question it answers
1	Purpose and scope	Who and what does it cover, and why?
2	Approved tools	Which tools, for what? Who approves new ones?
3	Data classification	Which data class (Public / Internal / Confidential / Restricted) may go into which tool?
4	Verification and review	Who checks AI-assisted work, how, and where is it recorded?
5	Disclosure	When do we tell clients, reviewers, or officials? How are AI images labeled?
6	Training	Who is trained, when, and on what?
7	Accountability and responsible charge	Who is responsible — up to the engineer who seals?
8	Incident reporting	What must be reported, to whom, and how fast?
9	Review cycle	When is the policy updated as tools and laws change?

Debate setup: what is shop-drawing review?

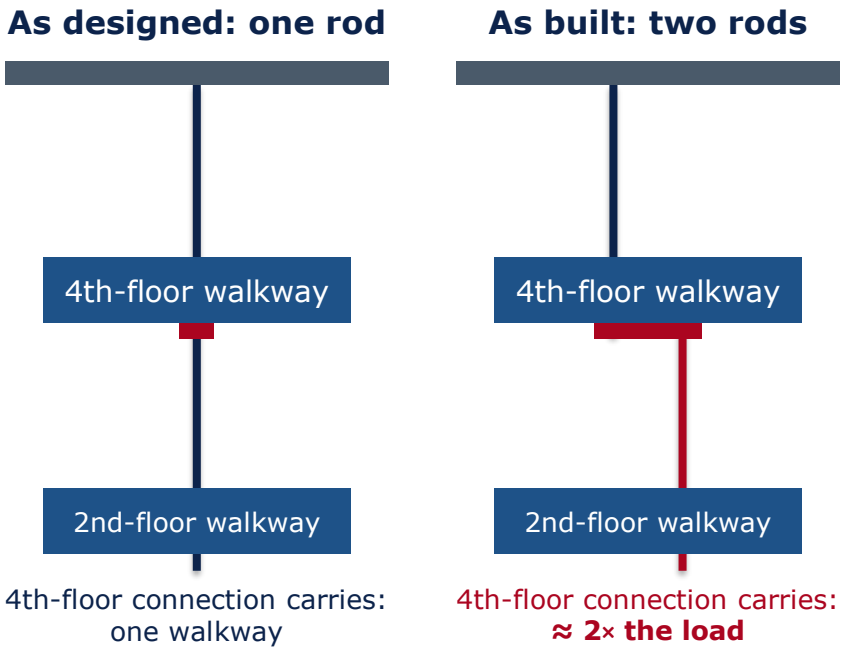
Shop drawings and submittals

Detailed drawings a fabricator or contractor makes to show how a component will be built — steel connections, rebar placement, precast panels, ductwork.

The engineer of record (EOR) reviews them only for **general conformance** with the design — not to approve or seal them — and returns an action stamp.

- Typical stamps: No exceptions taken · Make corrections noted · Revise and resubmit · Rejected
- Delegated design: a specialty engineer designs and seals that part; the EOR still reviews it

Hyatt Regency, 1981: a change in the shop-drawing process



- **17 July 1981, Kansas City:** two hotel walkways fell; 114 people died, 216 were injured
- The original connection already had only ~60% of the code-required capacity
- The fabricator's double-rod change doubled the load; it was accepted by phone, without calculations
- Missouri revoked the licenses of the EOR, the project engineer, and the firm

Shop-drawing review is part of the EOR's job — under **responsible charge**

Firm policy and debate

Debate: should an AI tool do the EOR's shop-drawing review?

12 min

Structured debate

- Your side of the room is your side: **YES** or **NO** — regardless of what you believe
- 3 min: prepare in groups. Use the codes, board rules, the EU AI Act, and Week 13 risks
- 2 min opening per side, then 1 min rebuttal per side
- Last 2 min: vote — then propose a **middle-ground rule** a firm could adopt

Motion: Should an AI tool perform the EOR's shop-drawing review and return the action stamp? **Consider:** public safety · competence · responsible charge · liability · error rates of people vs. AI · Hyatt Regency

Firm policy and debate

Debrief: AI can help check; the review stays with a person

AI as a checking aid

- Flags mismatches: sizes, dimensions, missing welds or bars
- Scans every sheet; people tire after hundreds
- The reviewer still decides the stamp and logs AI findings
- Fits current practice if verified and disclosed

vs

AI as the reviewer

- No one is in responsible charge of the review
- One flaw can repeat across a whole project
- Submittal review is the EOR's professional service — it can't be handed to a tool
- Liability and accountability become unclear

Middle ground: AI may help check; a responsible engineer reviews and returns the stamp

Lab 14

Lab day: AI policy workshop + project work

LAB

14

Bring a charged laptop

Open the notebook from the course GitHub page

- Teams draft a **one-page AI-use policy** for a small CE/ARCE firm from the nine-part template
- Test it on **5 scenarios** — e.g., an AI-drafted spec that cites "ASCE 7-19," an edition that does not exist
- Swap with another team and find one gap in their policy
- Second half: final-project work session with instructor check-ins
- **Lab 14 and HW5 are due Sunday 11:59 pm (Arizona time)**

Wrap-up

Key takeaways

1. AI rules come in layers: laws, state board rules, standards, society codes, organization policies — and your judgment
2. The EU AI Act regulates by risk; AI used as a safety component of critical infrastructure is **high-risk**
3. The U.S. (as of 2026): no single AI law — sector regulators, state laws, voluntary NIST and ISO frameworks
4. Your license follows your state board's rules; the NSPE and ASCE codes agree — **the engineer, not the tool, is responsible**
5. A firm AI policy has nine parts, from scope and data classes to incident reporting and a review cycle

Learn more

Readings and resources

- **EU AI Act — high-level summary** — about 15 min; Annex III lists the high-risk areas
- **NSPE Code of Ethics for Engineers** — the canons (10 min)
- **Arizona Rules of Professional Conduct, A.A.C. R4-30-301** — item 16 is the sealing rule; official text at btr.az.gov
- **ASCE Code of Ethics** — five stakeholders
- **ASCE Policy Statement 573: AI and Engineering Responsibility** — one page
- **NIST AI Risk Management Framework** — overview page
- **U.S. Copyright Office — Copyright and AI** — Part 2 (2025) summary

Appendix A1 · pre-reading

Who owns AI output — and whose data went in?

© Copyright needs a human author

U.S. Copyright Office (2025): material generated wholly by AI is not protected; prompts alone usually are not enough. Human selection, arrangement, and edits can be.

? Training data is contested

Courts are still deciding when training AI on copyrighted work is fair use — unsettled as of 2026.

§ Contracts decide a lot

A tool's terms of service and your client contract set who owns outputs and how your data may be used. Read them.

! Client data stays confidential

Drawings, reports, site photos, and bids belong to the client. Use only approved tools with data protections.

Appendix A2 · pre-reading

What may you paste where? Example data rules

	Public chatbot (free/personal account)	Enterprise AI (firm-licensed, data contract)	Local model (only on a firm computer)
Public — codes, published papers	✓ Yes	✓ Yes	✓ Yes
Internal — firm templates, draft calcs	✕ No	✓ Yes	✓ Yes
Confidential (client) — drawings, geotech reports, bids	✕ No	? Ask	? Ask
Restricted — personal data, security details	✕ No	✕ No	? Ask

Example only; your firm's policy and client contracts decide. ✓ Yes = allowed · ? Ask = only with written project-manager approval and if the client contract allows · ✕ No = not allowed

Appendix A3 · pre-reading

Our course's AI-use labels mirror professional practice

At UA, the Code of Academic Integrity covers AI use. When in doubt, ask.

PROVE YOUR OWN COMPETENCE

AI-Free

- **In this course:** quizzes, exam, exit tickets
- **In practice:** the FE and PE exams; any task that shows what you can do

NORMAL ENGINEERING WORK

AI-Assisted

- **In this course:** labs, most homework, the project
- **In practice:** tools allowed — you verify, disclose, and seal only what you checked

TESTING THE TOOL IS THE JOB

AI-Required

- **In this course:** HW4; model-testing parts of Assist and Evaluate projects
- **In practice:** piloting an AI tool before your firm adopts it